

Best-contacting ligand pose per state, backbone width and colour give each residue's contact frequency over all bound frames

Ligand atoms  
 ● C ● N ● O ● S

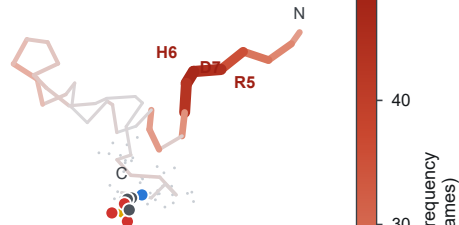
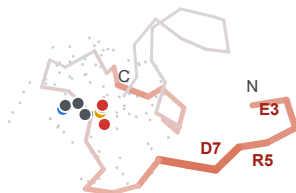
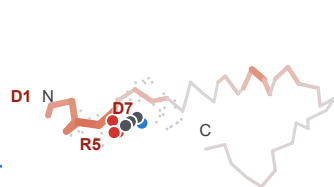
State 0

State 1

State 2

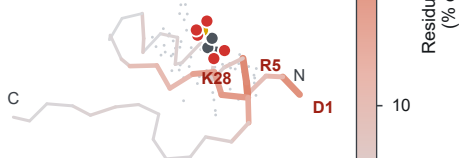
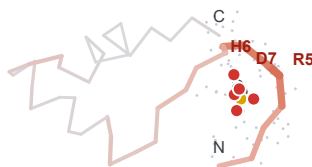
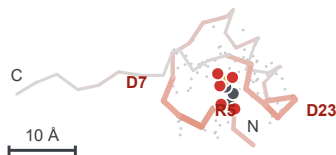
a

Aβ42 + TMP



b

Aβ42 + SPA



Residue-ligand contact frequency  
 (% of ligand-bound frames)

c

tramiprosate,  $C_3H_9NO_3S$   
 zwitterion: one cationic amine



SPA,  $C_3H_4O_5S$   
 dianion: no cationic group



**Why the two ligands pick different residues**

TMP carries a positive amine as well as a sulfonate, so it pairs with acidic side chains (Asp, Glu). SPA has no cationic group, so it can only salt-bridge to basic side chains (Arg, Lys). Fig. 3 shows exactly that split.